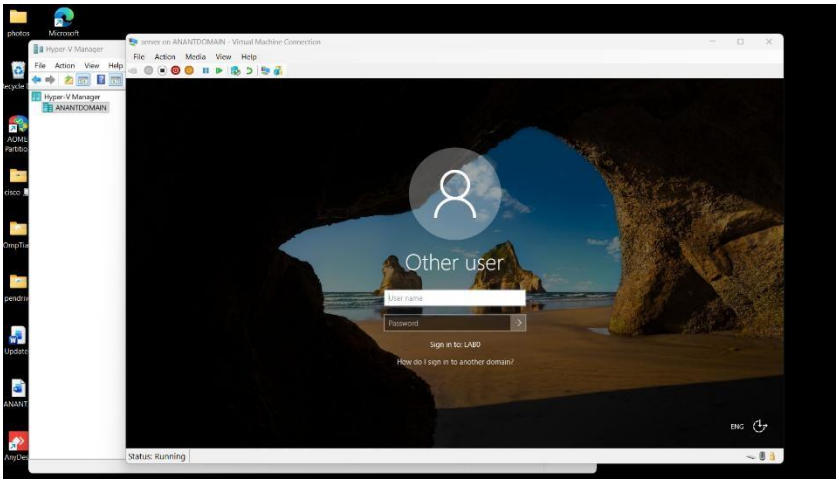
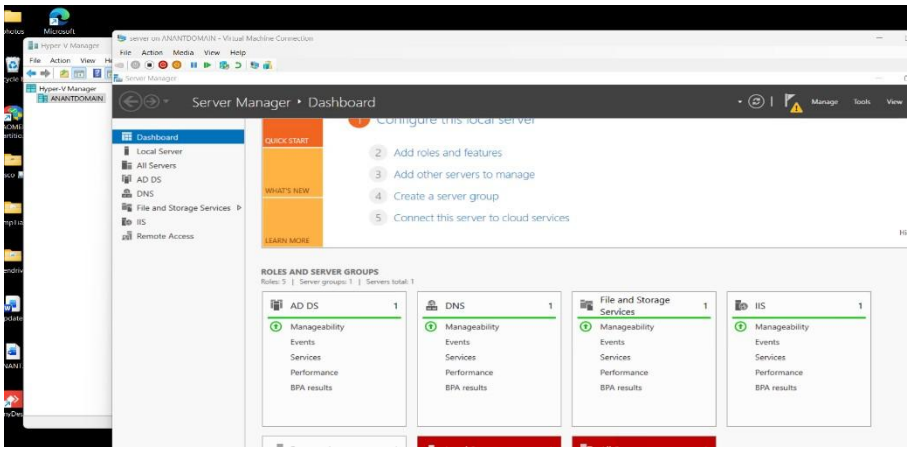
1. **Install Windows Server on a virtual machine using VirtualBox or VMware and take a screenshot of the desktop after the installation completes.**

    1. Open the virtualization software.
    2. Create a new virtual machine.
    3. Allocate RAM and create a virtual hard disk.
    4. Attach the Windows Server 2019 ISO file.
    5. Start the virtual machine and begin the installation.
    6. Select the Windows Server edition and complete the installation.
    7. Restart the virtual machine after installation.
    8. Log in using the Administrator account.
    9. Verify that the Windows Server desktop opens successfully.

Screenshots :



Windows Server 2019 Log in screen After Successful Installation.



Windows Server 2019 Virtual Machine Running in Hyper-V.

2. Open Server Manager and use the 'Add roles and features' wizard to install the Active Directory Domain Services (AD DS) role. Document each step with a short description and screenshot.

Step 1: Open Server Manager
Log in to Windows Server and open Server Manager. It is used to manage server roles, features, and configurations.

Step 2: Select Add Roles and Features
Click **Manage** and select **Add Roles and Features** to start the installation wizard.

Step 3: Before You Begin
Read the introductory information and click **Next**.

Step 4: Select Installation Type
Choose **Role-based or feature-based installation** and click **Next**.

Step 5: Select Destination Server
Select the local server from the server pool and click **Next**.

Step 6: Select Server Role
Check **Active Directory Domain Services (AD DS)** in the Server Roles list.

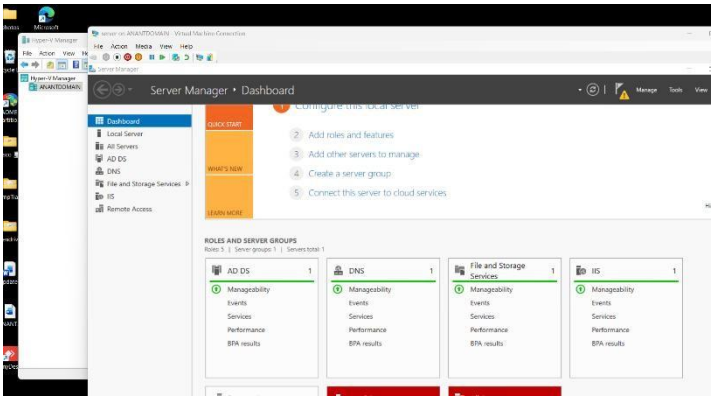
Step 7: Add Required Features
When prompted, click **Add Features** to include the required supporting features, then click **Next**.

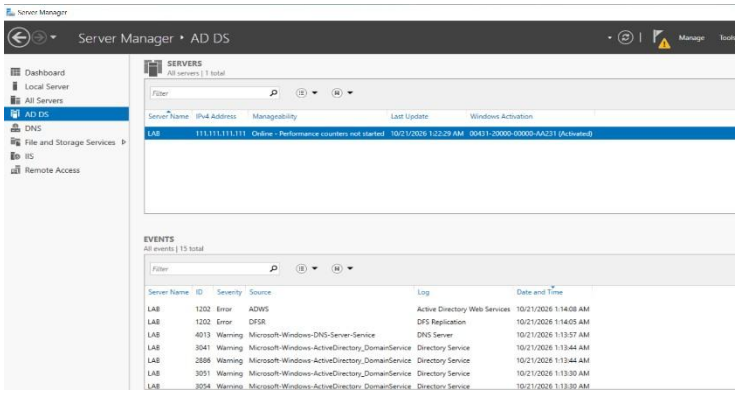
Step 8: Select Features
Keep the default selections unless additional features are required, then click **Next**.

Step 9: AD DS Information
Read the information about Active Directory Domain Services and click **Next**.

Step 10: Confirm Installation
Review the selected roles and features, then click **Install**.

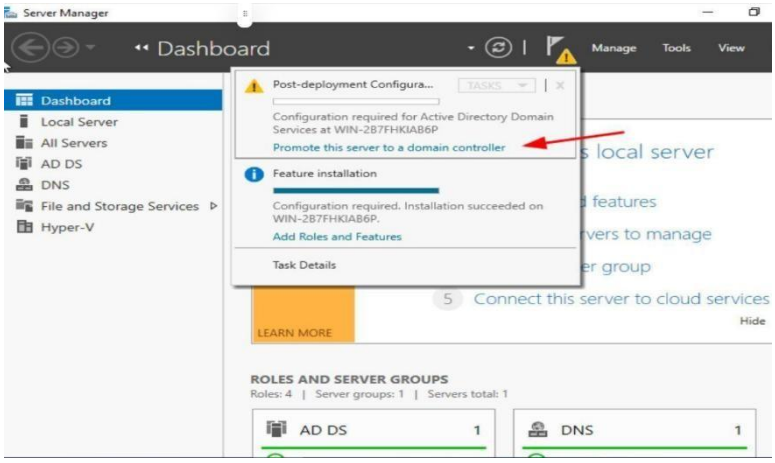
Step 11: Verify Installation
Wait for the installation to complete and verify that Active Directory Domain Services is installed successfully

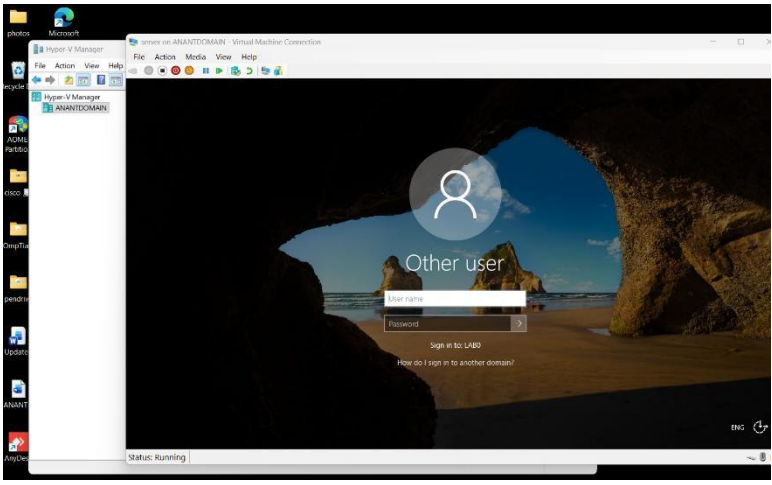




3. Promote your Windows Server to a Domain Controller by creating a new forest with a root domain name of your choice (e.g., mycollege.local). List the options you selected during the configuration.

1. Open Server Manager.
2. Click the notification flag and select **Promote this server to a domain controller**.
3. In Deployment Configuration, select **Add a new forest**.
4. Enter `anant.local` as the root domain name.
5. On the Domain Controller Options page, select the appropriate forest and domain functional levels supported by the server.
6. Keep **Domain Name System (DNS) server** and **Global Catalog (GC)** selected.
7. Enter and confirm the Directory Services Restore Mode (DSRM) password.
8. Review the DNS Options page and proceed.
9. Verify the NetBIOS domain name, usually `ANANT`.
10. Keep the default paths for the Active Directory database, log files, and SYSVOL unless a different location is required.
11. Review the configuration options and click **Next**.
12. Allow the prerequisite checks to complete. If no blocking errors are reported, click **Install**.
13. Wait for the server to restart automatically.
14. Log in using the domain Administrator account and verify that the server has become a Domain Controller.





4. **Create two new user accounts and one group in Active Directory Users and Computers. Assign both users to the group and describe how you did it.

Hint: Use the 'Users' container and right-click options in ADUC.**

Step 1: Open Active Directory Users and Computers
Open Server Manager, click Tools, and select Active Directory Users and Computers.

Step 2: Open the Users Container
Expand the domain `anant.local` and select the built-in Users container.

Step 3: Create the First User
Right-click the Users container, select New → User, and enter the user's name and logon name. Click Next, set a password, configure the password options, and click Finish.

Step 4: Create the Second User
Repeat the same procedure to create the second user account.

Step 5: Create a Group
Right-click the Users container, select New → Group, and enter the group name. Select Global as the group scope and Security as the group type, then click OK.

Step 6: Add Both Users to the Group
Right-click the newly created group and select Properties. Open the Members tab and click Add. Enter both user names, click Check Names, and then click OK.

Step 7: Verify Group Membership
Open the group's Properties and check the Members tab to confirm that both user accounts are listed.

